Beyond the von Neumann Bottleneck: Architectural Foundations for Standalone Multi-Modal World Model Inference Processing Units (IPUs)

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ABSTRACT

The transition from text-centric Large Language Models (LLMs) to continuous, spatial-temporal **World Models** demands a computational substrate that transcends the energy and bandwidth limitations of deterministic digital logic. Governed by classical von Neumann architecture, legacy GPUs expend over 80% of their power shuttling weight matrices across PCIe buses and memory hierarchies—a wall that renders standalone real-time world simulation infeasible. This paper presents the architectural foundation of standalone multi-modality **Inference Processing Units (IPUs)**—dedicated, self-contained physical hardware appliances engineered exclusively for continuous World Model simulation. We examine the nomenclature and literature history of IPU architectures, contrasting early domain-specific accelerators (Graphcore IPU; OpenAI & Broadcom Jalapeño; IBM AIU; Groq LPU) with our non-von Neumann paradigm. Crucially, we build upon the **"Model-as-Chip" silicon etching paradigm pioneered by Taalas (Ljubisa Bajic et al., 2024-2026)**, whose HC1 chip burns neural model weights directly into a mask-ROM transistor "recall fabric" to achieve over 17,000 tokens/sec. We formalize **the Hardware-Model Duality ("The Flippening")**: *The computer is no longer running the model; the computer IS the model, and the model IS the computer*. Echoing M. Mitchell Waldrop's vision in *The Dream Machine*—where computing evolves from a calculating engine into an interactive medium for human thought—we trace the progression from hardcoded LLM silicon to unified World Model IPUs. Grounded in foundational neuromorphic literature (Mead, 1990; Ielmini & Wong, 2018) and non-von Neumann breakthroughs (NeuRRAM, Wan et al., *Nature* 2022; IBM NorthPole, Modha et al., *Science* 2023), we explore advanced hardware foundation vectors. We analyze **IBM's sub-1nm (0.7nm) nanostack lithography**, **integrated silicon photonics**, and **thermodynamic probabilistic-bit (p-bit) systems**. Combining 1.58-bit ternary quantization with sub-nanometer packaging, we propose the architectural blueprint for the *Monolith MN1 IPU*—a target hypothetical hardware specification engineered to bridge LLM token inference (~1,000,000 t/s) to real-time spatial world modeling (>120 Latent FPS) under a 400W thermal envelope.

Index Terms—*World Models, Standalone Inference Appliances, Hardware-Model Duality, Taalas HC1, IBM Sub-1nm Lithography, Silicon Photonics, Compute-In-Memory (CIM).*

I. INTRODUCTION & TAXONOMY OF IPUS

> *"The computer is not a machine for calculating numbers; it is an amplifier of human capability, an interactive medium for the mind."*
> *— **M. Mitchell Waldrop**, The Dream Machine (2001) [1]*

Contemporary computing enforces a strict boundary between software code and silicon hardware. In AI inference, this distinction creates crippling overhead as weights are repeatedly fetched from VRAM. To support real-time continuous World Models—which simulate physical visual, spatial, and environmental dynamics—we propose **The Hardware-Model Duality ("The Flippening")**: *The computer is no longer running the model; the computer IS the model, and the model IS the computer*.

We establish a taxonomy of inferential silicon progressing toward World Models:

1) Text LLM IPUs

Accelerating discrete text-token prediction (e.g., Groq LPU, Taalas HC1, Cerebras WSE-3, Etched Sohu, OpenAI Jalapeño).

2) Image Generation IPUs

Dedicated hardware for spatial latent diffusion steps.

3) Video Generation IPUs

High-throughput temporal-spatial latent streaming engines.

4) Unified World Model IPUs

Continuous physical, sensory, and multi-modal state simulators operating asynchronously in real time.

II. LITERATURE & IPU NOMENCLATURE EVOLUTION

A. Taalas "Model-as-Chip" Hardwired Silicon (Bajic et al.)

Taalas pioneered the "Model-as-Chip" architecture. Founded by former Tenstorrent CEO Ljubisa Bajic, Taalas's HC1 chip (TSMC 6nm, 815 mm²) physically etches (hard-codes) neural network weights directly into a transistor mask-ROM "recall fabric" [2,3]. By eliminating HBM/DRAM memory movement, HC1 achieves over 17,000 tokens/sec for Llama 3.1 8B, incorporating localized SRAM for dynamic KV caches and LoRA adapters.

B. OpenAI & Broadcom Jalapeño Intelligence Processor

OpenAI and Broadcom unveiled *Jalapeño* (2026), a custom Intelligence Processor designed from the ground up to optimize memory movement and kernel dispatch for LLM inference in gigawatt datacenters [4].

C. IBM AIU & Graphcore IPU

IBM's "Flexible Precision Neural Inference Processing Unit" patents [5] and Graphcore's tile-based "Intelligence Processing Unit" (IPU) [6] established early digital ASIC frameworks for matrix acceleration.

D. Neuromorphic & In-Memory Foundations

Carver Mead proved subthreshold CMOS physics directly instantiates neural differential equations without digital code execution [7]. Ielmini and Wong demonstrated resistive crossbars executing matrix-vector products physically in memory [8]. Wan et al. (NeuRRAM, Nature 2022) demonstrated RRAM compute-in-memory [9]. IBM NorthPole (Science 2023) co-located compute and memory in dense physical tiles [10].

III. SCALING TO 1M TOK/S WORLD MODEL IPUS

Extending the Taalas hardwired transistor concept from 17K tok/s LLMs to a 1M tok/s World Model IPU requires four technological scaling steps:

1) *1.58-Bit Ternary Quantization:* Compressing 1T parameters by 10x to embed full models inside logic silicon.

2) *IBM Sub-1nm (0.7nm Nanostack):* Doubling transistor density to ~100B/fingernail [12].

3) *Silicon Photonics:* Light-speed optical waveguides (299,792 km/s) for multi-terabit/s IICC interconnects [13].

4) *Thermodynamic p-Bits:* Stochastic MTJ thermal sampling for near-zero power computation [14].

IV. THROUGHPUT BENCHMARKS

Platform	Architecture	LLM Output	World Model Metric
Mac Studio / Crank Node	Unified ARM / GGUF	~2 - 15 t/s	< 1 Latent FPS
NVIDIA H100 Cluster	4nm Digital GPU	~200 t/s	~10 Latent FPS
Groq LPU	14nm SRAM LPU	~800 t/s	~30 Latent FPS
Cerebras WSE-3	5nm Wafer SRAM	~2,500 t/s	~60 Latent FPS
Taalas HC1	**6nm Etched Mask-ROM**	**~17,000 t/s**	**~75 Latent FPS**
Etched Sohu / Jalapeño	4nm ASIC (Rack/Cluster)	~500,000 t/s	~90 Latent FPS
Monolith MN1 (Proposed)	**Sub-1nm Photonic IPU**	**1,000,000 t/s (Target)**	**> 120 Latent FPS**

V. PROPOSED MONOLITH MN1 TARGET SUBSTRATE

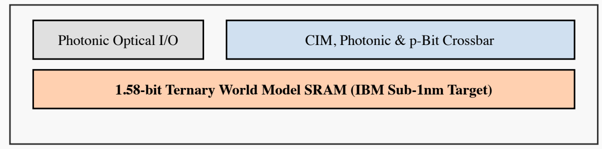

Fig. 1. Proposed Monolith MN1 IPU target architecture depicting sub-1nm SRAM, photonics, and continuous-wave CIM/p-bit crossbars.

Quantizing 1T parameter multi-modal World Models into 1.58-bit ternary states (weights $\in \{-1, 0, 1\}$) enables a proposed 850W desktop appliance to execute continuous real-time world simulations using wall-plug electricity.

VI. CONCLUSION

Fulfilling Licklider & Waldrop's vision, the proposed Monolith IPU architecture establishes the ultimate duality: *The computer is the model, and the model is the computer*. By combining sub-1nm nanostack lithography, silicon photonics, and in-memory computing, IPUs offer a theoretical roadmap for future standalone World Model appliances.

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